

## UMNUGOVI, Mongolia

WMO: 442150

Lat: 49.1047N

Lon: 91.7261E

Elev: 5217

StdP: 12.13

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -34.7      | -29.4 | -38.8                         | 0.7 | -34.3 | -33.4 | 1.0 | -28.7 | 27.6                  | 11.8 | 22.0 | 13.1 | 0.6                   | 270  | 0.674 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.4                   | 80.9            | 58.6 | 77.9 | 57.2 | 75.1 | 56.2 | 61.5                | 75.9 | 59.7 | 73.1 | 58.2 | 70.9 | 8.2                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 56.6                            | 82.9 | 65.9 | 54.7 | 77.2 | 64.7 | 52.9 | 72.2 | 63.9 | 30.2          | 76.4 | 28.8 | 72.9 | 27.6 | 71.0 | 73.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 26.0          | 21.0 | 18.7 | DB                         | -34.4 | 85.8               | 8.0 | 2.9                                                | -40.2 | 87.9       | -44.9 | 89.6       | -49.4 | 91.2       | -55.2 | 93.3 | (4) |
| (5)               |      |      | WB                         | -34.3 | 64.0               | 8.0 | 3.4                                                | -40.0 | 66.5       | -44.7 | 68.4       | -49.2 | 70.3       | -55.0 | 72.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 31.8   | -8.2  | -0.1  | 19.8  | 37.7  | 49.5  | 60.7  | 64.2  | 59.9  | 47.8  | 33.3  | 15.0  | -0.4  |  |  |
|                                                                        | DBStd       | 26.53  | 11.68 | 13.28 | 12.05 | 9.32  | 8.13  | 6.05  | 5.04  | 5.94  | 7.83  | 8.89  | 11.03 | 11.25 |  |  |
|                                                                        | HDD50       | 7902   | 1805  | 1402  | 937   | 378   | 109   | 6     | 0     | 5     | 130   | 518   | 1050  | 1562  |  |  |
|                                                                        | HDD65       | 12221  | 2270  | 1822  | 1402  | 820   | 482   | 150   | 74    | 175   | 517   | 982   | 1500  | 2027  |  |  |
|                                                                        | CDD50       | 1248   | 0     | 0     | 0     | 9     | 93    | 328   | 441   | 313   | 63    | 1     | 0     | 0     |  |  |
|                                                                        | CDD65       | 91     | 0     | 0     | 0     | 0     | 1     | 22    | 50    | 18    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 823    | 0     | 0     | 0     | 1     | 21    | 215   | 402   | 173   | 11    | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 115    | 0     | 0     | 0     | 0     | 1     | 24    | 67    | 22    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 5.4    | 2.6   | 3.4   | 6.0   | 7.8   | 7.8   | 6.7   | 5.8   | 5.1   | 5.5   | 5.7   | 4.6   | 3.5   |  |  |
|                                                                        | PrecAvg     | 5.1    | 0.0   | 0.0   | 0.1   | 0.2   | 0.3   | 0.9   | 1.6   | 1.2   | 0.4   | 0.1   | 0.1   | 0.1   |  |  |
|                                                                        | PrecMax     | 9.8    | 0.2   | 0.1   | 0.3   | 0.7   | 1.3   | 2.7   | 4.3   | 2.6   | 1.5   | 0.8   | 0.5   | 0.5   |  |  |
| (15) Precipitation                                                     | PrecMin     | 2.8    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 1.7    | 0.0   | 0.0   | 0.1   | 0.2   | 0.3   | 0.7   | 0.9   | 0.7   | 0.4   | 0.2   | 0.1   | 0.1   |  |  |
|                                                                        | DB          | 26.9   | 34.0  | 54.0  | 66.6  | 77.0  | 83.1  | 86.2  | 83.1  | 75.1  | 60.7  | 48.0  | 33.1  |       |  |  |
|                                                                        | MCWB        | 21.1   | 25.6  | 39.5  | 48.6  | 53.6  | 58.2  | 60.7  | 59.4  | 54.2  | 45.1  | 38.2  | 25.9  |       |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 20.4   | 28.8  | 48.2  | 61.8  | 71.9  | 79.2  | 81.7  | 78.6  | 70.2  | 55.9  | 41.6  | 27.4  |       |  |  |
|                                                                        | 2% MCWB     | 16.3   | 22.2  | 36.1  | 45.1  | 52.0  | 56.6  | 59.7  | 58.0  | 51.8  | 42.8  | 33.5  | 22.0  |       |  |  |
|                                                                        | 5% DB       | 15.1   | 24.2  | 43.7  | 57.6  | 68.1  | 76.2  | 78.4  | 75.2  | 66.2  | 51.9  | 36.1  | 22.2  |       |  |  |
|                                                                        | 5% MCWB     | 11.7   | 18.8  | 33.2  | 42.6  | 49.9  | 55.9  | 58.2  | 56.9  | 50.0  | 40.1  | 29.2  | 17.8  |       |  |  |
|                                                                        | 10% DB      | 9.7    | 19.4  | 38.9  | 53.8  | 64.5  | 73.3  | 75.6  | 72.3  | 62.5  | 48.2  | 31.3  | 17.4  |       |  |  |
|                                                                        | 10% MCWB    | 7.3    | 15.1  | 30.3  | 40.4  | 48.2  | 54.7  | 57.0  | 55.5  | 48.4  | 37.6  | 25.5  | 13.9  |       |  |  |
|                                                                        | WB          | 21.3   | 26.1  | 40.8  | 50.5  | 55.9  | 62.6  | 64.7  | 63.1  | 56.7  | 46.6  | 38.6  | 26.2  |       |  |  |
|                                                                        | MCDB        | 26.3   | 33.0  | 51.9  | 63.2  | 71.9  | 78.1  | 78.9  | 77.2  | 72.9  | 58.0  | 46.9  | 33.0  |       |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 16.3   | 22.4  | 36.8  | 46.1  | 53.6  | 59.8  | 62.3  | 60.3  | 53.2  | 43.7  | 33.5  | 22.0  |       |  |  |
|                                                                        | 2% MCDB     | 20.3   | 28.7  | 47.5  | 59.3  | 69.0  | 74.0  | 77.2  | 73.6  | 67.4  | 54.9  | 41.0  | 27.4  |       |  |  |
|                                                                        | 5% WB       | 11.8   | 18.9  | 33.5  | 43.8  | 51.5  | 57.8  | 60.5  | 58.5  | 51.1  | 40.7  | 29.4  | 17.8  |       |  |  |
|                                                                        | 5% MCDB     | 15.0   | 24.1  | 43.0  | 56.2  | 65.9  | 72.0  | 74.2  | 71.2  | 64.2  | 50.9  | 35.8  | 22.1  |       |  |  |
|                                                                        | 10% WB      | 7.2    | 15.3  | 30.4  | 41.3  | 49.3  | 56.0  | 58.9  | 57.0  | 49.0  | 37.9  | 25.7  | 14.0  |       |  |  |
|                                                                        | 10% MCDB    | 9.6    | 19.3  | 38.6  | 53.0  | 62.9  | 69.9  | 71.4  | 69.3  | 61.1  | 47.5  | 31.1  | 17.4  |       |  |  |
|                                                                        | MDBR        | 20.2   | 24.4  | 24.9  | 22.4  | 21.1  | 19.7  | 18.4  | 19.2  | 22.3  | 21.7  | 19.7  | 18.5  |       |  |  |
|                                                                        | 5% DB       | 23.0   | 26.7  | 27.0  | 26.4  | 26.1  | 24.2  | 23.1  | 23.6  | 27.2  | 26.4  | 22.9  | 20.7  |       |  |  |
| (36) Mean Daily Temperature Range                                      | MCWBR       | 19.5   | 21.2  | 17.9  | 15.1  | 13.6  | 10.4  | 9.5   | 9.6   | 13.6  | 16.5  | 16.3  | 16.8  |       |  |  |
|                                                                        | MCDBR       | 23.0   | 26.7  | 26.0  | 24.9  | 23.3  | 22.2  | 20.8  | 21.3  | 25.1  | 25.3  | 22.3  | 20.7  |       |  |  |
|                                                                        | MCWBR       | 19.5   | 21.4  | 17.7  | 15.4  | 13.3  | 11.1  | 10.0  | 10.1  | 13.5  | 16.6  | 16.1  | 16.8  |       |  |  |
|                                                                        | MCDBR       | 23.0   | 26.7  | 26.0  | 24.9  | 23.3  | 22.2  | 20.8  | 21.3  | 25.1  | 25.3  | 22.3  | 20.7  |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.235  | 0.262 | 0.307 | 0.353 | 0.351 | 0.356 | 0.353 | 0.334 | 0.297 | 0.279 | 0.255 | 0.235 |       |  |  |
|                                                                        | taud        | 2.342  | 2.285 | 2.225 | 2.204 | 2.265 | 2.335 | 2.373 | 2.402 | 2.455 | 2.429 | 2.359 | 2.315 |       |  |  |
|                                                                        | Ebn at Noon | 270    | 285   | 286   | 283   | 288   | 286   | 285   | 287   | 289   | 276   | 257   | 252   |       |  |  |
|                                                                        | Edh at Noon | 21     | 28    | 36    | 42    | 41    | 38    | 36    | 34    | 29    | 24    | 20    | 18    |       |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 494    | 830   | 1304  | 1757  | 2084  | 2139  | 2067  | 1831  | 1478  | 937   | 524   | 384   |       |  |  |
|                                                                        | RadStd      | 25     | 35    | 64    | 64    | 121   | 91    | 115   | 74    | 59    | 52    | 43    | 26    |       |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           |  | N/A   | N/A     | -5.49  | N/A     | -1.26 | -2.36 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (7 neighbors) |  | N/A   | N/A     | -4.25  | N/A     | -1.62 | -2.23 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page